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# SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

**Table 1: Rigor Adherence Table**

| |
|---|
| Ethics |
| not required. |
| Inclusion and Exclusion Criteria |
| not required. |
| Attrition |
| not required. |
| Sex as a biological variable |
| not required. |
| Subject Demographics |
| Age: not required. |
| Weight: not required. |
| Randomization |
| not detected. |
| Blinding |
| not detected. |
| Power Analysis |
| not detected. |
| Replication |
| not required. |

**Table 2: Key Resources Table**

| Your Sentences | REAGENT or RESOURCE | SOURCE | IDENTIFIER |
|---|---|---|---|
| Software and Algorithms | | | |
| The second strategy was the profile-profile comparison method which is implemented in the HHpred program 56. | HHpred | | Suggestion: HHpred (RRID:SCR_010276)( link) |
| 2.3 Sequence similarity-based clustering For the sequences collected from the previous series of searches, similarity based clustering was performed to eliminate highly similar sequences by using CD-HIT, based on various sequence identity thresholds (0.4-0.7), and the BLASTCLUST program which adjusts the length of aligned regions and bit-score density threshold empirically (ftp:ftp.ncbi.nih.govblastdocumentsblastclust.html). | CD-HIT | | Suggestion: CD-HIT (RRID:SCR_007105)( link) |
| | BLASTCLUST | | Suggestion: BLASTClust (RRID:SCR_016641)( link) |
| The revised collection of the representative sequences from all major globin families was subjected to all-against-all BLASTP comparisons, and clustered based on the sequence similarity using the force-directed graph drawing (Fruchterman and Reingold) algorithm 57 implemented in the CLANS program 58. | BLASTP | | Suggestion: BLASTP (RRID:SCR_001010)( link) |
| The alignment was colored using an in-house alignment visualization program written in Perl and further modified using Adobe Illustrator. | Adobe Illustrator | | Suggestion: Adobe Illustrator, RRID:SCR_010279( link) |
| Structure comparisons were conducted by using the DALI program, which generates an optimal pairwise structural alignment based on the similarity of local patterns extracted from contact maps 63. | DALI | | |
| Other structural analysis operations were conducted using the molecular visualization program PyMOL 64. | PyMOL | | Suggestion: PyMOL, RRID:SCR_000305( link) |
| We detected protein domains using the HMMER package which searches against the Pfam database and a local collection of profile HMMs. | HMMER | | |
| | Pfam | | |
| We determined the conserved co-occurring genes based on 1) the frequency of common domains identified in the neighborhood, 2) the presence of such co-occurring genes in at least two phylogenetically | NCBI Taxonomy | | |

| | | | |
|---|---|---|---|
| distinct lineages (phylum in NCBI Taxonomy database), and 3) complete conservation in a particular lineage (phylum). | | | |

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq